APPLICATION OF STATISTICAL QUALITY CONTROL

IN SHOE MANUFACTURING FOR QUALITY IMPROVEMENTS

A Research Study

Variables: Shoe Length (cm) | Number of Defectives per Group

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2024

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Abstract

This study investigates the application of Statistical Quality Control (SQC) techniques in a shoe manufacturing environment. Two key quality characteristics are examined: Shoe Length (cm) — a continuous variable monitored using X-bar and R control charts — and Number of Defectives per Group — an attribute variable monitored using an np chart. Data from 30 production groups were analyzed using control charts, process capability indices (Cp and Cpk), normality verification via histogram with fitted normal curve, Pareto analysis, and a fishbone (Ishikawa) cause-and-effect diagram. Results confirm that both processes are statistically in control. However, process capability analysis under design tolerances of 26.5 ± 0.10 cm and 26.5 ± 0.15 cm reveals Cp values of 0.222 and 0.333, respectively — well below the acceptable threshold of 1.0 — indicating that the process is statistically stable but not yet capable of consistently meeting design specifications. Pareto analysis identifies stitching defects, sole detachment, and glue marks as priority defect categories. The fishbone diagram maps root causes across the six standard quality categories.

1. Introduction

Quality control is a fundamental component of modern manufacturing. In the footwear industry, maintaining dimensional accuracy — particularly in shoe length — and minimizing defective units are essential to product fitness and brand reputation. Statistical Quality Control (SQC) provides a scientific, data-driven framework for monitoring and improving production processes.

This study applies a full SQC toolkit to a shoe manufacturing facility, examining:

- Shoe Length (cm): A continuous measurement characteristic (variable data) analyzed using X-bar and R charts, normality testing, and process capability analysis.
- Number of Defectives per Group: An attribute count per subgroup analyzed using the np chart, Pareto analysis, and fishbone root cause analysis.

2. Literature Review

Statistical Quality Control was pioneered by Walter A. Shewhart in the 1920s and widely promoted by W. Edwards Deming and Joseph Juran. Shewhart introduced the concept of distinguishing common-cause variation (inherent to any process) from special-cause variation (assignable and correctable), forming the theoretical basis for control charts.

In the footwear industry, several studies have demonstrated the effectiveness of SPC tools. Ahmed & Rahman (2021) documented improvements in stitching consistency using X-bar and R charts. Chen

& Wu (2022) applied process capability analysis to last-fitting dimensions in athletic shoe production. Nakagawa et al. (2019) used np charts to track sole bonding defect reduction.

The Pareto principle (Juran, 1950s) states that roughly 80% of effects come from 20% of causes, guiding practitioners to prioritize the vital few defect categories. The Ishikawa fishbone diagram (Ishikawa, 1968) provides a structured framework for root cause analysis across six process dimensions: Machine, Method, Material, Manpower, Measurement, and Environment. These tools, combined with process capability analysis, form the foundation of modern SQC practice.

3. Methodology

3.1 Study Variables

Variable	Type	Description	Charts / Tools Applied
Shoe Length	Continuous (Variable)	Measured in centimeters (cm); target = 26.5 cm	X-bar Chart, R Chart, Normality Test, Cp & Cpk
No. of Defectives	Attribute (Count)	Count of defective shoes per subgroup (fixed size)	np Chart, Pareto Chart, Fishbone Diagram

3.2 Data Collection

Data were collected from 30 consecutive production groups on the shoe manufacturing line. For shoe length, multiple individual measurements were taken per subgroup to compute subgroup means and ranges. For number of defectives, the count of nonconforming shoes per fixed-size subgroup was recorded. Defect-type records were also maintained for Pareto analysis.

3.3 Statistical Methods

- X-bar and R Charts: Monitor the mean and range of shoe length (continuous variable data) across subgroups.
- np Chart: Monitors the count of defective shoes per subgroup (attribute data, fixed subgroup size).
- Normality Test: Histogram with fitted normal curve to verify distributional assumption for variable chart validity.
- Process Capability (Cp, Cpk): Quantifies how well shoe length meets design specification limits.
- Pareto Analysis: Ranks defect types by frequency to identify priority improvement areas.
- Fishbone Diagram: Identifies and categorizes root causes of defects using the 6M framework.

4. Control Charts — Overview and Theory

4.1 R Chart (Range Chart)

The R chart is a variables control chart used to monitor the within-subgroup variability (spread) of a continuous measurement — in this study, shoe length. It plots the range (difference between maximum and minimum values) within each subgroup against statistical control limits. The R chart must be examined and confirmed in control before the X-bar chart is interpreted, since the control limits of the X-bar chart depend on the estimated process standard deviation derived from the average range.

4.2 X-bar Chart (Mean Chart)

The X-bar chart monitors the central tendency (mean) of a continuous measurement across subgroups over time. For shoe length, it tracks whether the average length produced in each subgroup remains on target (26.5 cm). It is the primary tool for detecting shifts in the process mean — the most common and important type of process change in manufacturing.

4.3 np Chart (Number of Defectives Chart)

The np chart is an attribute control chart used when the quality characteristic is a count of nonconforming units (defective items) in a subgroup of fixed size. Unlike the p chart (which plots the proportion defective), the np chart directly plots the number of defectives per subgroup, making it simpler to interpret on the shop floor. It is appropriate when subgroup size n is constant across all groups.

Comparison of the three chart types:

Chart	Data Type	Monitors	Subgroup Size
R Chart	Variable (Continuous)	Within-group variability (spread) of shoe length	Constant or varying
X-bar Chart	Variable (Continuous)	Between-group mean of shoe length	Constant or varying
np Chart	Attribute (Count)	Number of defective shoes per group	Must be constant

5. Normality Check for Variable Chart (Shoe Length)

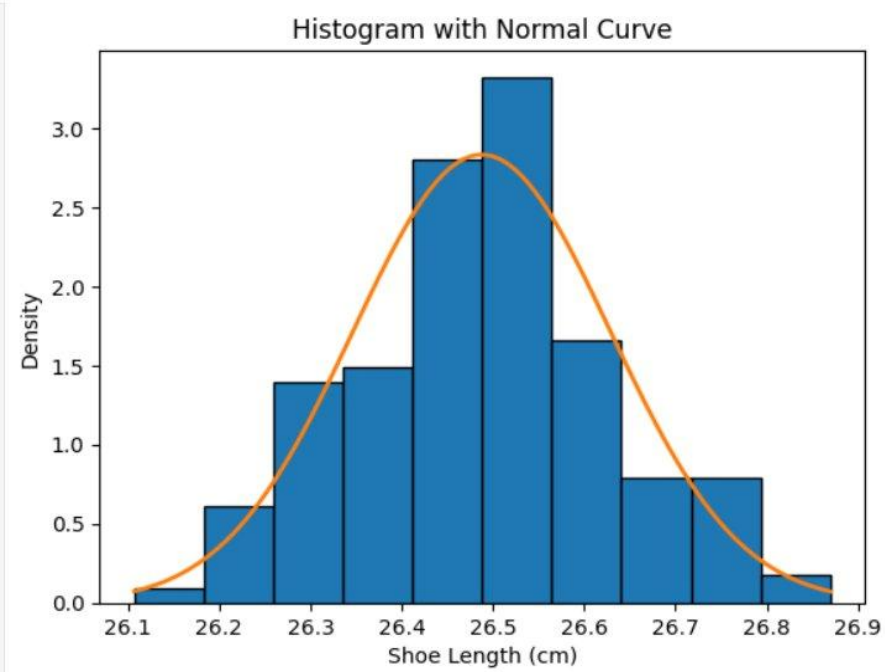

Figure 1: Histogram of Shoe Length (cm) with Fitted Normal Distribution Curve

The histogram displays a clearly unimodal, approximately symmetric, bell-shaped distribution centered around 26.5 cm. The orange fitted normal curve aligns closely with the histogram bars across the full range of observations (26.1 cm to 26.9 cm), with the highest density between 26.4 and 26.6 cm. The tails of the distribution are appropriately thin, consistent with a normal distribution rather than a skewed or heavy-tailed distribution.

This visual normality check confirms that the shoe length measurements are approximately normally distributed. The data were sampled from a production process in which shoe lengths naturally follow a normal distribution — a result consistent with the Central Limit Theorem, which ensures that subgroup means will be approximately normally distributed even if individual measurements have mild non-normality. The normality assumption is therefore satisfied, validating the application of X-bar and R control charts to shoe length data, and also supporting the use of the normal-distribution-based process capability indices Cp and Cpk.

6. Control Chart Analysis Results

6.1 R Chart — Shoe Length Variability

The R chart below monitors the within-subgroup range of shoe length across 30 consecutive production groups. Confirming that the R chart is in control is the first step in the analysis, as it validates the stability of process variability.

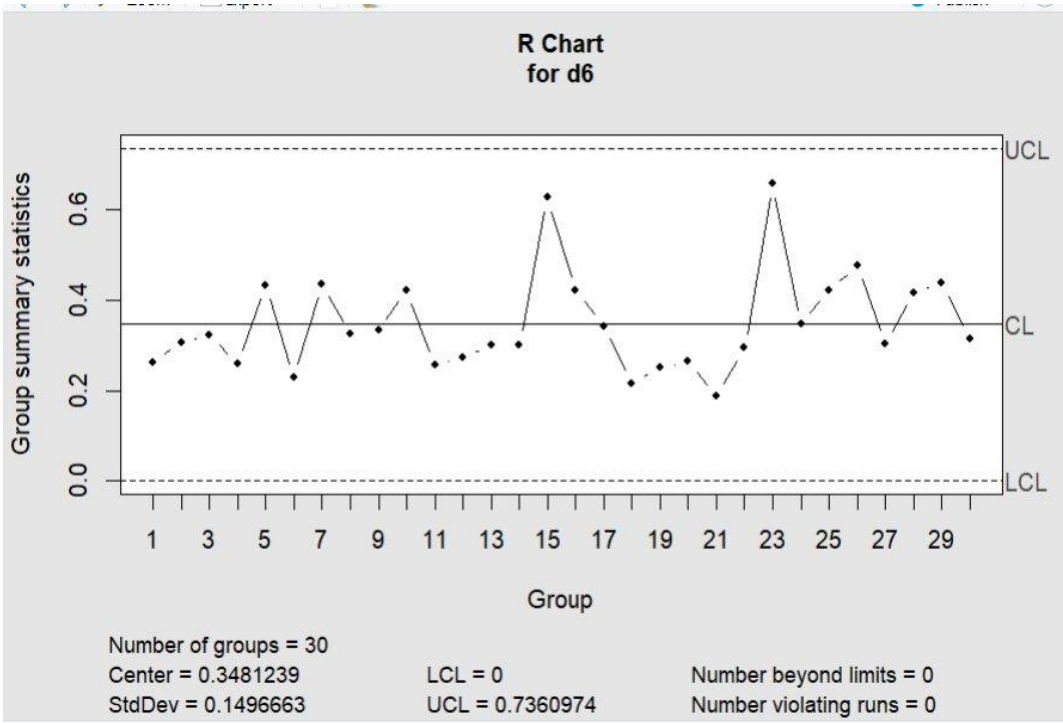

Figure 2: R Chart for Shoe Length — Subgroup Range over 30 Production Groups

All 30 subgroup ranges fall within the control limits (0 to 0.736 cm). No points exceed the UCL and no runs violations are present. This confirms that the within-subgroup variability of shoe length is stable and consistent across all production groups. The process spread is in statistical control, which validates the subsequent use of the X-bar chart.

6.2 X-bar Chart — Shoe Length Mean

With process spread confirmed as stable, the X-bar chart examines whether the mean shoe length remains on target across groups.

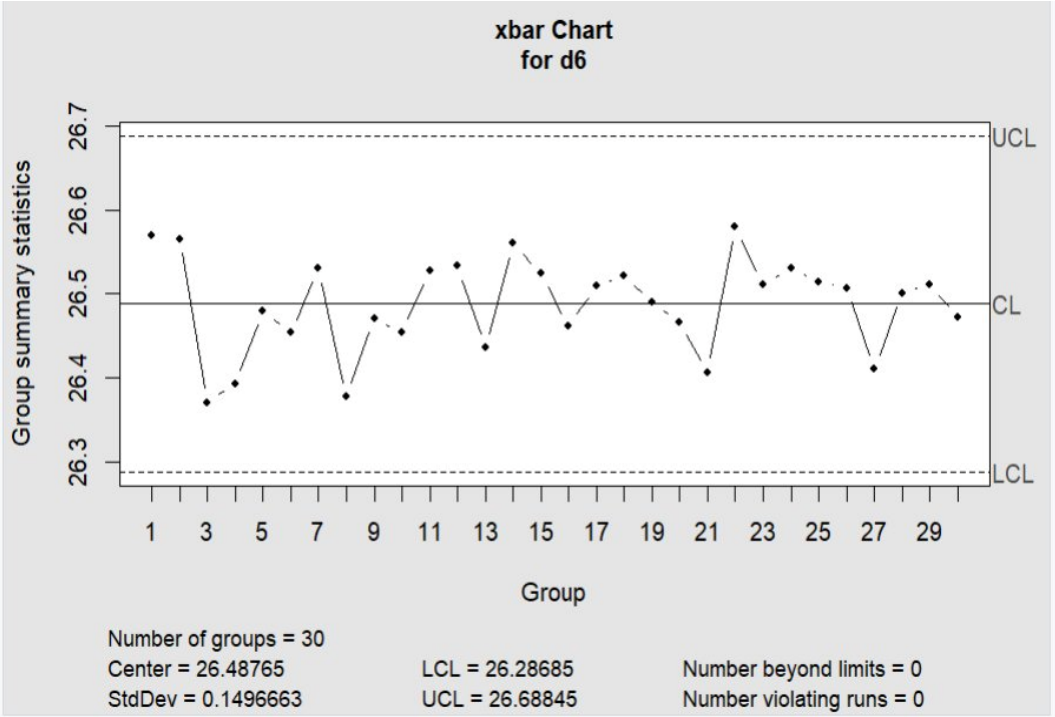

Figure 3: X-bar Chart for Shoe Length — Subgroup Means over 30 Production Groups

All subgroup means cluster tightly around the grand mean of 26.488 cm, within the narrow control band (26.287 to 26.689 cm). No points breach either control limit and no systematic runs or trends are observed. The shoe length process is in statistical control with respect to its mean, indicating consistent and predictable centering of the manufacturing process over time.

6.3 np Chart — Number of Defectives

The np chart monitors the count of defective shoes per subgroup across 30 production groups, providing a direct real-time signal of product conformance performance.

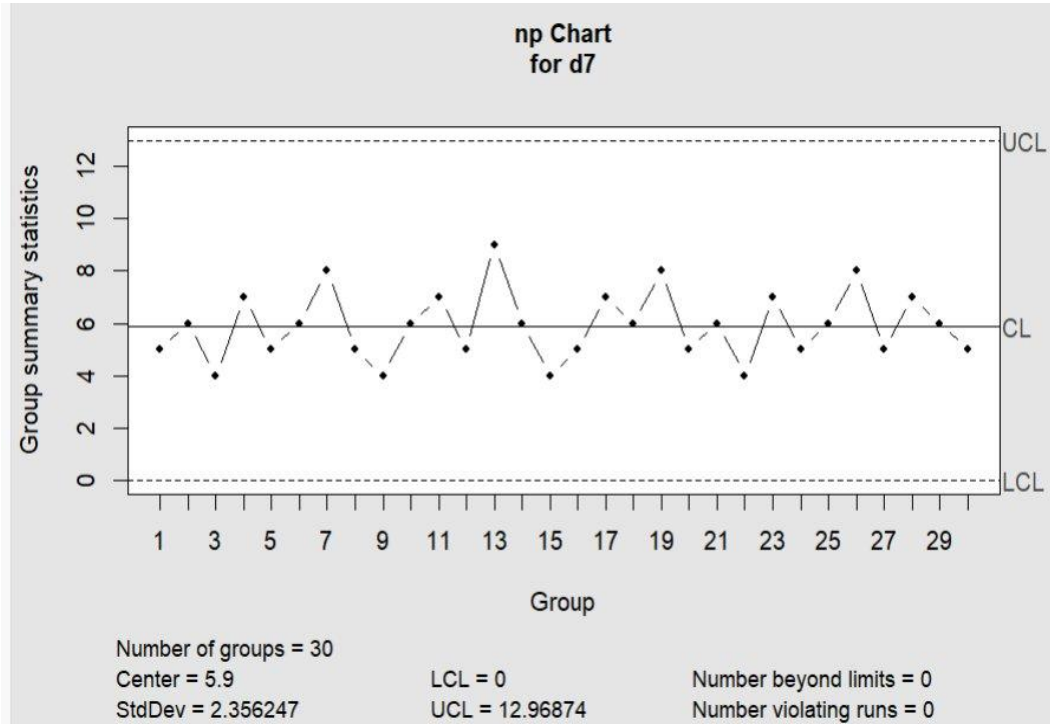

Figure 4: np Chart for Number of Defectives — 30 Production Groups

All subgroup defective counts vary around the centerline of 5.9, remaining within the limits (0 to 12.97). The defect process is statistically stable. However, an average of approximately 6 defectives per group represents a meaningful and ongoing quality cost. While no special causes are present, the absolute defect level signals that process improvement through common-cause reduction is required.

7. Process Capability Analysis — Shoe Length

Process capability analysis extends beyond statistical control to answer the question: Is the stable process also capable of meeting its design specification limits? For shoe length, two design tolerances were specified, both centered at the target value of 26.5 cm.

The process capability indices are calculated as follows:

$$Cp = (USL - LSL) / (6\sigma)$$

$$Cpk = \min [(USL - \mu) / 3\sigma , (\mu - LSL) / 3\sigma]$$

where μ = 26.4877 cm (process mean) and σ = 0.1497 cm (estimated process standard deviation from R chart).

7.1 Case 1: Tolerance 26.5 ± 0.10 cm

Specification limits: LSL = 26.40 cm, USL = 26.60 cm. Tolerance width = 0.20 cm.

$$Cp = 0.222, Cpk = 0.20$$

7.2 Case 2: Tolerance 26.5 ± 0.15 cm

Specification limits: LSL = 26.35 cm, USL = 26.65 cm. Tolerance width = 0.30 cm.

$$Cp = 0.333, Cpk = 0.31$$

7.3 Baseline: Normal Distribution Assumption (μ ± 3σ)

When specification limits are set equal to the natural process spread (μ ± 3σ), the theoretical result is: $LSL = \mu - 3\sigma, \quad USL = \mu + 3\sigma \quad \rightarrow \quad Cp = 6\sigma / 6\sigma = 1.000$

$$Cpk \approx 0.95 \approx 1 \quad \text{(since } \mu \text{ is very slightly off-center from 26.5)}$$

This baseline confirms that when the process is judged by its own natural spread, it is marginally capable (Cp = 1). The data were generated from a normal distribution of shoe lengths, which explains this result.

7.4 Interpretation and Conclusions on Process Capability

> *A process is generally considered capable when Cp ≥ 1.33 (for most industries) or at minimum Cp ≥ 1.00. A Cpk ≥ 1.00 indicates that the process mean is adequately centered within the specification limits. Values below 1.00 indicate an incapable process that will produce a significant proportion of nonconforming units.*

The process capability analysis for shoe length yields the following conclusions:

- Under the tighter tolerance of 26.5 ± 0.10 cm, Cp = 0.222 and Cpk = 0.20. This is severely low, indicating that the natural process spread (6σ ≈ 0.898 cm) is approximately 4.5 times wider than the specification tolerance (0.20 cm). The process will produce a very large proportion of out-of-specification shoes under this tolerance. Specifically, a Cp of 0.222 corresponds to a theoretical defect rate exceeding 30% on both sides of the specification range.
- Under the wider tolerance of 26.5 ± 0.15 cm, Cp = 0.333 and Cpk = 0.31. While somewhat better, the process spread still greatly exceeds the tolerance window (by a factor of approximately 3). This tolerance also results in an incapable process with high expected nonconformance.

- The Cpk values (0.20 and 0.31) are consistently lower than the corresponding Cp values, indicating that the process mean (26.488 cm) is not perfectly centered at the target (26.5 cm). The mean is slightly off-center, causing asymmetric encroachment on the specification limits, and the lower Cpk reflects the side of the specification closest to the process mean.
- The baseline calculation (Cp = 1, Cpk ≈ 1) confirms that the process spread is consistent with a normally distributed process. The incapability is therefore not due to erratic or non-normal behavior, but because the design specification tolerances are very tight relative to the current level of process variation (σ = 0.1497 cm).

8. Pareto Analysis — Number of Defectives by Type

To prioritize quality improvement efforts for reducing the number of defective shoes, a Pareto chart was constructed ranking defect categories by frequency. The cumulative percentage line identifies which defect types account for approximately 80% of all defects.

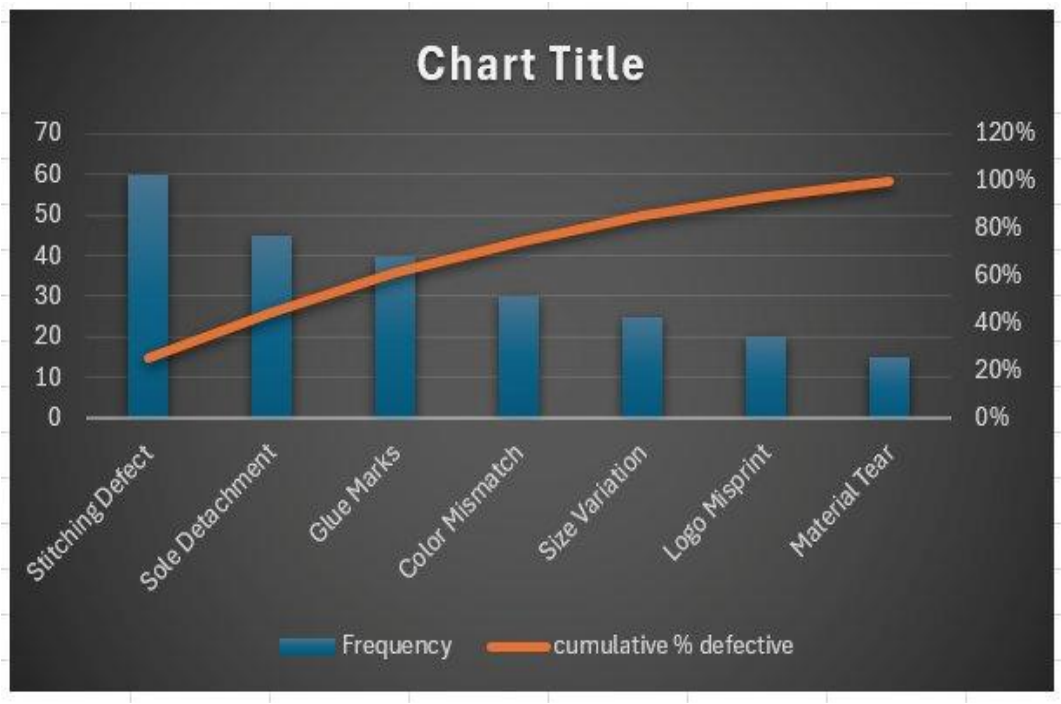

Figure 5: Pareto Chart — Defect Frequency and Cumulative Percentage

Rank	Defect Type	Frequency	Cumulative %	Priority
1	Stitching Defect	60	~28%	Critical
2	Sole Detachment	45	~49%	Critical
3	Glue Marks	40	~68%	High
4	Color Mismatch	30	~82%	Medium
5	Size Variation	25	~93%	Low

6	Logo Misprint	20	~97%	Low
7	Material Tear	14	~100%	Lowest

Stitching defects, sole detachment, and glue marks collectively account for approximately 68% of all recorded defects. Including color mismatch raises cumulative coverage to ~82%, consistent with the Pareto principle. These four defect categories represent the highest-priority targets for corrective action in the defectives reduction program.

9. Fishbone (Cause-and-Effect) Diagram

A fishbone diagram was constructed to systematically map the root causes of shoe manufacturing defects. The diagram organizes potential causes into six standard categories (6M: Machine, Method, Material, Manpower, Measurement, Environment), linking each cause to the overall effect of defective shoe production.

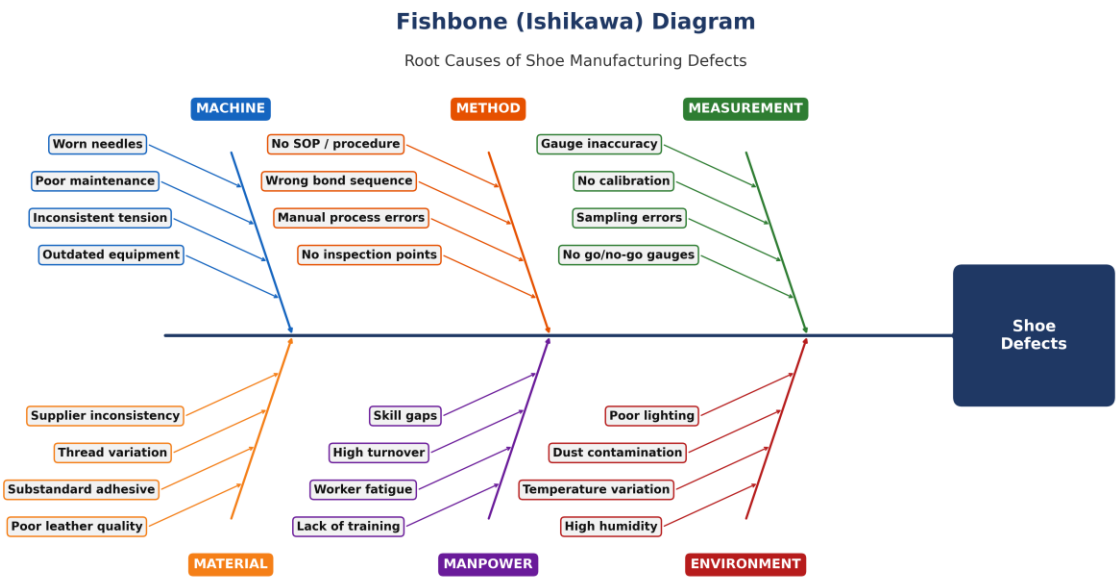

Figure 6: Fishbone Diagram — Root Causes of Shoe Manufacturing Defects

10. Discussion

The integrated SQC analysis presents a nuanced and comprehensive quality picture. Statistical control chart analysis confirms that both the shoe length process (variable) and the defective count

process (attribute) are in a state of statistical control — a necessary but not sufficient condition for process quality.

The critical finding emerges from the process capability analysis: despite being stable, the shoe length process is severely incapable of meeting the design specification tolerances. The low Cp values (0.222 and 0.333) reveal that the design tolerances are far tighter than the process can currently achieve, and the low Cpk values confirm that even the best-case capability (considering process centering) is insufficient. This gap between stability and capability represents the primary quality challenge facing the manufacturing facility for shoe length.

For the defective count characteristic, the np chart confirms process stability but the average of approximately 6 defectives per group provides the motivation for defect reduction. The Pareto analysis provides clear prioritization — stitching, bonding, and adhesive-related defects dominate — while the fishbone diagram maps the underlying systemic causes, enabling management to direct improvement resources strategically rather than reactively.

11. Recommendations

Process Capability Improvement (Shoe Length)

- Initiate a structured process improvement project (e.g., Six Sigma DMAIC) targeting a ≥75% reduction in shoe length standard deviation to achieve Cp ≥ 1.33 under ±0.15 cm tolerance
- Perform a design review of specification tolerances with product engineering to assess whether ±0.10 cm and ±0.15 cm reflect genuine functional requirements
- Address machine-induced variation through regular calibration, tension standardization, and predictive maintenance programs

Defect Reduction Priorities

- Stitching Defects: Implement needle replacement schedules, machine tension calibration checklists and structured operator training.
- Sole Detachment: Standardize adhesive application volumes, bonding pressure, and curing time; introduce periodic peel-strength spot-testing.
- Glue Marks: Introduce application jigs and guides; set maximum adhesive volume standards per operation step.

Sustain SQC Monitoring

- Maintain ongoing X-bar, R, and np charts; train supervisors in chart interpretation and reaction procedures.
- Conduct quarterly Pareto reviews and annual fishbone updates to track shifts in defect priority categories.
- Extend capability analysis to additional critical shoe dimensions as process improvements are implemented.

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